

6. Project Testing

Objective

The objective of testing is to verify that the application performs correctly under different input conditions and accurate recommendations and predictions..

Test Environment

Operating System:

Windows / Linux / macOS

IDE:

Visual Studio Code

Programming Language:

Python

Framework:

Flask

Test Cases

Test Case 1 — Valid Recommendation

Input:

User submits valid Nitrogen, Phosphorous, Potassium, Temperature, Humidity, Soil pH and Rainfall via "Recommend crop."

Expected Result:

A recommended crop with a confidence percentage is displayed, and SoilData, Crop, Prediction and Report rows are all created and correctly linked.

Actual Result:

"Recommendation ready — rice · 99.57%" displayed; database confirmed matching soil_data, crop, prediction (with soil_id/crop_id/model_id set) and report rows.

Status:

PASS

Test Case 2 — Missing Values

Input:

Form submitted with all fields left empty.

Expected Result:

A validation error listing each missing field is shown and no recommendation is generated.

Actual Result:

“Request failed — Nitrogen (N) is required. · Phosphorous (P) is required. · Potassium (K) is required. · Temperature (°C) is required. · Humidity (%) is required. · Soil pH is required. · Rainfall (mm) is required.”

Status:

PASS

Test Case 3 — Out-of-Range Values

Input:

Form submitted with values outside each field's allowed range.

Expected Result:

A validation error listing the allowed range for each field is shown and no recommendation is generated.

Actual Result:

“Request failed — Nitrogen (N) must be between 0 and 200. · Phosphorous (P) must be between 0 and 200. · Potassium (K) must be between 0 and 220. · Temperature (°C) must be between -5 and 55. · Humidity (%) must be between 0 and 100. · Soil pH must be between 0 and 14. · Rainfall (mm) must be between 0 and 400.”

Status:

PASS

Performance Testing

The prediction response, including SoilData/Crop/Report writes, is generated within a few seconds.

Result

The application passed all functional and security tests successfully.

Conclusion

Testing confirmed that system performs reliably and produces accurate and suitable crop recommendations and predictions.